

Practical MATLAB Modeling with Simulink

**Programming and Simulating
Ordinary and Partial Differential
Equations**

Sulaymon L. Eshkabilov

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Practical MATLAB Modeling with Simulink: Programming and Simulating Ordinary and Partial Differential Equations

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Printed on acid-free paper

To the memory of my father.

To my mother.

To my wife, Nigora, after 25 wonderful years together.

Table of Contents

About the Author xiii

About the Technical Reviewerxv

Acknowledgmentsxvii

Introductionxix

Part I: Ordinary Differential Equations..... 1

Chapter 1: Analytical Solutions for ODEs..... 3

 Classifying ODEs 4

 Example 1 5

 Example 2 6

 Example 3 6

 Analytical Solutions of ODEs 8

 dsolve() 8

 Example 4 8

 Example 5 9

 Example 6 11

 Example 7 11

 Second-Order ODEs and a System of ODEs 13

 Example 8 13

 Example 9 14

 Example 10 15

 Example 11 16

 Example 12 16

 Example 13 17

 Laplace Transforms..... 22

TABLE OF CONTENTS

Example 14 24

 laplace/ilaplace 25

Example 15 25

Example 16 26

Example 17 26

Example 18 26

Example 19 30

Example 20 34

Example 21 36

References..... 40

Chapter 2: Numerical Methods for First-Order ODEs..... 41

 Euler Method..... 41

 Example 1 42

 Improved Euler Method..... 44

 Backward Euler Method..... 45

 Example 2 47

 Midpoint Rule Method 50

 Example 3 51

 Ralston Method 55

 Runge-Kutta Method 56

 Example 4 57

 Runge-Kutta-Gill Method 60

 Runge-Kutta-Fehlberg Method 63

 Adams-Bashforth Method 66

 Example 5 66

 Milne Method 72

 Example 6 73

 Taylor Series Method 75

 Example 7 75

 Adams-Moulton Method..... 78

Example 8	80
MATLAB's Built-in ODE Solvers	85
Example 9	87
The OPTIONS, ODESET, and ODEPLOT Tools of Solvers	93
Example 10	95
Example 11	98
Simulink Modeling	103
Example 12	103
SIMSET	111
References	112
Chapter 3: Numerical Methods for Second-Order ODEs	113
Euler Method	114
Example 1	114
Example 2	118
Example 3	120
Example 4	122
Example 5	124
Runge-Kutta Method	128
Example 6	128
Example 7	131
Example 8	134
Example 9	136
Example 10	138
Adams-Moulton Method	141
Example 11	141
Example 12	146
Simulink Modeling	151
Example 13	151
Example 14	153
Example 15	156

TABLE OF CONTENTS

Example 16 157

Nonzero Starting Initial Conditions 158

Example 17 159

ODEx Solvers..... 162

Example 18 163

Example 19 165

Example 20 166

Example 21 169

diff() 172

Example 22 172

Chapter 4: Stiff ODEs 177

Example 1 177

Example 2 179

Example 3 180

Example 4 183

Jacobian Matrix 185

Example 5 186

Example 6 189

Chapter 5: Higher-Order and Coupled ODEs..... 193

Fourth-Order ODE Problem 193

Robertson Problem 197

Akzo-Nobel Problem 199

HIRES Problem 206

Reference..... 211

Chapter 6: Implicit ODEs..... 213

Example 1 214

Example 2 218

Example 3 220

Example 4 223

Example 5	226
Example 6	229
References.....	232
Chapter 7: Comparative Analysis of ODE Solution Methods	233
Example 1	234
Drill Exercises	251
Exercise 1	251
Exercise 2	252
Exercise 3	252
Exercise 4	253
Exercise 5	253
Exercise 6	254
Exercise 7	255
Exercise 8	256
Exercise 9	257
Exercise 10	258
Exercise 11	258
Exercise 12	259
Exercise 13	259
Part II: Boundary Value Problems in Ordinary Differential Equations	261
Chapter 8: Boundary Value Problems	263
Dirichlet Boundary Condition Problem	267
Example 1	267
Example 2	271
Robin Boundary Condition Problem	274
Example 3	274
Sturm-Liouville Boundary Value Problem	279
Example 4	280
Stiff Boundary Value Problem	285

TABLE OF CONTENTS

Example 5 285

References 289

Drill Exercises 289

Exercise 1 290

Exercise 2 290

Exercise 3 290

Exercise 4 290

Exercise 5 290

Exercise 6 291

Exercise 7 291

Exercise 8 291

Exercise 9 291

Exercise 10 291

Exercise 11 292

Exercise 12 292

Exercise 13 292

Part III: Applications of Ordinary Differential Equations..... 293

Chapter 9: Spring-Mass-Damper Systems 295

Single Degree of Freedom System..... 295

Case 1: Free Vibration (Motion) 295

Case 2: Forced Vibration (Motion) 309

Two Degrees of Freedom System 320

Three Degrees of Freedom System..... 328

Matrix Approach for n -Degree of Freedom System 336

References 343

Chapter 10: Electromechanical and Mechanical Systems..... 345

Modeling a DC Motor 345

Modeling a DC Motor with Flexible Load 350

Modeling a Microphone 356

Modeling Motor: Pump Gear Box.....	361
Modeling Double Pendulum.....	371
References.....	384
Chapter 11: Trajectory Problems	385
Falling Object	385
Thrown Ball Trajectories	388
References.....	400
Chapter 12: Simulation Problems.....	401
Lorenz System	401
Lotka-Volterra Problem	407
References.....	411
Drill Exercises	412
Exercise 1	412
Exercise 2	413
Exercise 3	413
Exercise 4	413
Exercise 5	414
Exercise 6	414
Exercise 7	415
Exercise 8	415
Exercise 9	416
Exercise 10	416
Exercise 11	416
Exercise 12	417
Exercise 13	417
Exercise 14	418
Exercise 15	419
Exercise 16	420
Exercise 17	422
Exercise 18	423

Part IV: Partial Differential Equations 427

Chapter 13: Solving Partial Differential Equations 429

 pdepe() 430

 One-Dimensional Heat Transfer Problem 431

 Example 1 432

 Two-Dimensional Heat Transfer: Solving an Elliptic PDE with the Gauss-Seidel Method 436

 Example 2 438

 del2(): Laplace Operator 443

 Example 3 444

 Wave Equation 447

 Solving a One-Dimensional Wave Equation 448

 Example 4 451

 Solving a Two-Dimensional Wave Equation 455

 Example 5 456

 References 459

 Drill Exercises 460

 Exercise 1 460

 Exercise 2 460

 Exercise 3 460

 Exercise 4 461

 Exercise 5 461

 Exercise 6 461

 Exercise 7 461

 Exercise 8 461

 Exercise 9 462

 Exercise 10 462

 Exercise 11 462

 Exercise 12 463

 Exercise 13 463

Index..... 465

About the Author



Dr. Sulaymon L. Eshkabilov is currently a visiting professor in the Department of Agriculture and Biosystems Engineering at North Dakota State University. He obtained his ME diploma from Tashkent Automobile Road Institute in 1994, his MSc from Rochester Institute of Technology, USA in 2004, and his PhD from Academy Sciences of Uzbekistan in 2005. He was an associate professor at Tashkent Automobile Road Institute from December 2006 to January 2017. He also held visiting professor and researcher positions at Ohio University from 2010 to 2011 and Johannes

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His research areas are mechanical vibrations, control, mechatronics, system dynamics, image processing, and microstructure analysis of materials. He is the author of more than 30 research papers and 5 books. Three of the five books are devoted to MATLAB/Simulink applications for mechanical engineering students and numerical analysis. From 2009 to 2020, he was an external academic expert for the European Commission, assessing academic projects.

About the Technical Reviewer



Karpur Shukla is a research fellow at the Centre for Mathematical Modelling at FLAME University in Pune, India. His current research interests focus on topological quantum computation, nonequilibrium and finite-temperature aspects of topological quantum field theories, and applications of quantum materials effects for reversible computing. He received an MSc in physics from Carnegie Mellon University, with a background in theoretical analysis of materials for spintronics applications as well as Monte Carlo simulations for the renormalization group of finite-temperature spin lattice systems.

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Introduction

This book covers the most essential and hands-on tools and functions of the MATLAB and Simulink packages and the Symbolic Math Toolbox (MuPAD notes) to solve, model, and simulate ordinary differential equations (ODEs) and partial differential equations (PDEs). It explains how to solve ODEs and PDEs symbolically and numerically via interactive examples and case studies. The main principle of the book is “learn by doing,” moving from the simple to the complex. This book contains dozens of solved problems and simulation models embedded in MATLAB scripts and Simulink models, which will help you to master programming and modeling essentials, as well as learn how to program and model more difficult and complex problems that involve ODEs and PDEs.

Practical MATLAB Modeling with Simulink explains various practical issues of programming and modeling in parallel by comparing the programming tools of MATLAB to the modeling tools of Simulink. By studying this book, you’ll be proficient at using the MATLAB/Simulink packages and at using the source codes and models from the book’s examples as templates for your own projects to solve modeling and simulation, or engineering problems with ODEs and PDEs.

What You Will Learn

By the end of the book, you’ll have learned how to do the following:

- Model complex problems using MATLAB and Simulink
- Use MATLAB and Simulink to solve ODEs and PDEs
- Use numerical methods to solve first-, second-, and higher-order and coupled ODEs
- Solve stiff and implicit ODEs
- Employ numerical methods to solve first- and second-order linear PDEs

INTRODUCTION

- Solve ODEs symbolically with the Symbolic Math Toolbox of MATLAB
- Understand the applications and modeling aspects of differential equations in solving various simulation problems

This book is aimed at engineers, programmers, data scientists, and students majoring in engineering, applied/industrial math, data science, and scientific computing. This book continues where Apress' *Beginning MATLAB and Simulink* leaves off.

The book is composed of three parts. Part 1 consists of the following chapters:

- Chapter 1 is dedicated to general formulations and solving ODEs symbolically using The Symbolic Math Toolbox functions and MuPAD note commands.
- Chapter 2 covers the most essential programming aspects to solve first-order ODEs numerically by writing scripts based on the Euler, Runge-Kutta, Milne, Adams-Bashforth, Ralston, and Adams-Moulton methods. You'll write the scripts using MATLAB's built-in ODE solvers, such as ode23, ode23t, ode45, ode15s, ode113, odeset, etc., and Simulink modeling.
- Chapter 3 is dedicated to solving second-order ODEs symbolically and numerically using the Euler, Runge-Kutta, Milne, Adams-Bashforth, Ralston, and Adams-Moulton methods. You'll do this using MATLAB's built-in ODE solvers, such as ode23, ode23t, ode45, ode15s, ode113, odeset, etc., and Simulink modeling.
- Chapter 4 addresses the issues of how to solve stiff ODEs by adjusting a step size, specifying a solver setting, selecting an appropriate solver type, and so forth.
- Chapter 5 is dedicated to solving higher-order and coupled ODEs.
- Chapter 6 is devoted to solving implicitly defined IVPs and differential algebraic equations using MATLAB's ode15i and Simulink modeling.

- Chapter 7 is dedicated to a comparative analysis of solving ODEs with MATLAB's ODE solvers, Simulink modeling, script writing, and computing analytical solutions with the Symbolic MATH Toolbox (MuPAD notes).

Part 2 consists of the following chapter:

- Chapter 8 covers most essential tools of solving boundary value problems of ODEs numerically by using MATLAB solvers such as `bvp4c`, `bvp5c`, `bvpinit`, `deval`, etc.

Part 3 consists of the following chapters:

- Chapter 9 is devoted to applications of ODEs, specifically, modeling and simulations of spring-mass-damper systems.
- Chapter 10 is devoted to applications of ODEs, specifically, modeling and simulations of mechanical and electromechanical systems.
- Chapter 11 is devoted to applications of ODEs, specifically, modeling and simulations of trajectory problems.
- Chapter 12 is devoted to applications of ODEs, specifically, modeling and simulations of several simulation problems, such as the Lotka-Volterra and Lorenz systems.

Part 4 consists of the following chapter:

- Chapter 13 is devoted to solving partial differential equations using PDE toolbox functions, such as `pdepe()` and the Laplacian operator `del2()`, and writing scripts based on Gauss-Seidel and finite difference methods.

How to Access the Source Code

All of the source code (such as M/MN files, Simulink models, and SLX/MDL files) discussed in the book is available to readers via the Download Source Code button at www.apress.com/9781484257982.

A Note to Users

The given scripts in the context of the book may not be the best solutions to the given problems, which was done intentionally in some cases to emphasize methods used to improve them; in other cases, the given scripts are the most appropriate solutions to the best knowledge of the author. Should I spot any better alternative solutions to exercises given in the context of the book, I intend to publish them via the MathWorks MATLAB Central User Community's file exchange via my file exchange link there—under my name.

No matter how hard we have worked to proofread the content of the book, it is inevitable that there might be some typographical errors that have slipped through and will appear in print. My apologies.

Sulaymon L. Eshkabilov
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